Mehmet Can Demir

Jr. Data Scientist

+90 533 152 7790 | mhmtcndmr01@gmail.com

Personal Website
LinkedIn Profile
Kaggle Profile
Github Profile

Education

Cukurova University: Computer Engineering **Degree**: Bachelor of Engineering **Sept 2018 – Sept 2022**

Work Experience

Data Science Intern **Jul 2023 – Dec 2023**

Doğuş Teknoloji *Remote*

- Built a Generative AI Assistant to reduce the HR Department's workload using **LLMs**, then deployed the model on **Azure**. I used **Streamlit (a python library)** for the frontend, users were able to select the subject and ask questions instead of reading all the related documents to find answers.
- Rewrote an entire ML Interface project that was written with **Python** and **Streamlit** as the frontend, with my new code architecture, more than 10 bugs were fixed and application speed increased. Then the new code **was proposed to the CEO** and some other **important possible customers**.
- Wrote a script to remove unnecessary tokens before processing, with this **Prompt Engineering** approach **LLMs** cost is **reduced by 30%**, this led to more cost-efficient processing.
- The team had a difficulty while scraping a specific website data, to solve this problem I proposed using **Regular Expressions (RegEx)**, and I scraped 600 pages of web data built with HTML and SVGs.
- I learned the **Azure AI** services interface and led the team on this IaaS by giving a small introduction to the team on how we can utilize it to use in our upcoming AI projects.

Data Science Intern **June 2022 – Aug 2022**

Jotform *Remote*

- Analyzed form creation data and applied **ZScore** statistical approach to detect outliers, in which case some of those outliers appeared as a fraud activity by some users. Used **Jupyter** as the software.
- To utilize Deep Learning and Anomaly Detection together, I made a presentation of which deep learning methods we can apply to detect anomalies, then, we can automate and create a smart alert system.
- Prepared 2 automated anomaly detection demos: **FacebookProphet** and **GANs**. While **FacebookProphet** showed a great performance, **GANs** needed more development and data.

Technical Skills

- **OS:** Windows, Arch Linux, Ubuntu
- **Languages:** C/C++, Python, SQL
- **Libraries:** TF, Keras, PyTorch, PySpark, Sklearn, Matplotlib, Numpy, Pandas, Seaborn, OpenCV
- **Tools:** Jupyter, Git, Linux, Docker, FastAPI, Django, Postman

Projects

U.S. Wildfire Analysis: Analyzed 1.88M wildfires in the U.S. and tried to predict the cause of the fires.

- Used **SQL** queries to get the desired specific data from the **SQLite database**.
- Written in **Python**. Used **Seaborn** as the data visualization tool.
- Predicting the possible cause of the fire is challenging, one of the reasons for this is the amount of the imbalanced data.
- Prepared a REPORT to answer the following questions about the fire frequency, most and least fire-prone counties and the predictability of the fire with given size, location and date.

WGAN-GP Trainer: Customizable Wasserstein Generative Adversarial Networks.

- Written with **Python**. Built networks using **Pytorch**. Records logs with **Tensorboard**.
- Uses 64x64 images to train and generate an image.
- Cross-platform, can be used in both Windows and Linux.
- Customizable: you can create your own architecture and use it instead of only 64x64.
- Includes DOCUMENTATION of how the project is made and info about the architecture.

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Volunteer Experience

ML Engineer (Freelance) **Sep 2021 – June 2022**
NutzenTech *Remote*

- Built Object Detection Models using both **PyTorch** and **TensorFlow** on local GPUS to be used in drones. The data used for training was gathered from a scientific paper.
- Drones were responsible for detecting defects and dirt on solar panels and wind turbines.
- Built a pipeline for object detection using **AWS SageMaker**, **S3 Bucket** and **Lambda**.
- Analyzed the amount of CO2 gas at certain spots and 3D visualized the amount of gas with the exact location and time using **Matplotlib**.

AI Engineer **Dec 2020 – Apr 2021**
Apziva *Remote*

- Analyzed candidate data using **Pandas**, **Matplotlib**, **Seaborn** data visualization tools on **Jupyter**, and presented to the team in as a detailed report about if we can predict a candidate's task submission.
- Trained a Random Forest classification model using **Scikit-Learn** to predict a candidate will or will not complete his task submission. Users' submission delay and profile data is used as an input, and the model was successful with an accuracy of 92%. This helped the HR department to track candidates.

ML Engineer **Sept 2020 – Apr 2021**
Cukurova Hydromobile Team, Cukurova University *Adana, Turkey*

- Built line detection algorithm using Hough Transform technique with **Python** and **OpenCV**.
- Improved line detection by applying image processing methods such as Gaussian Blurring, Canny Edge Detection and Greyscaling.
- Trained Object Detection models using **YOLOv4** and **TensorFlow Object Detection API** on Google Colab GPUs.
- Manually prepared a case-specific dataset that consists of over 1200 traffic signs and traffic lights images. The data was collected by taking photos of specific traffic signs on a sunny day. With the new dataset, the model has been re-trained and the accuracy **increased** to over **90%** from 70%.

R&D Engineer **May 2020 – Sep 2020**
CENGA R&D Team, Cukurova University *Adana, Turkey*

- Contributed a Deep Learning interface project by adding a **Scikit-Learn SVM** model training page using **Python** and **Tkinter GUI library**.
- Contributed to the team by finding specific research papers and patents, and summarizing their concepts and use-cases.

Languages

- Turkish (Native)
- English (Professional)
- German (Beginner)

External Links

Github Profile: https://github.com/mcandemir
LinkedIn Profile: https://www.linkedin.com/in/mehmet-can-demir/
Kaggle Profile: https://www.kaggle.com/mehmetcandemir
Personal Website: https://portfolio-mcandemir.vercel.app/

U.S. Wildfire Analysis: https://github.com/mcandemir/us-wildfire-analysis
REPORT: https://github.com/mcandemir/us-wildfire-analysis/blob/master/report.pdf

WGAN-GP Trainer: https://github.com/mcandemir/wgan-gp-trainer
DOCUMENTATION: https://github.com/mcandemir/wgan-gp-trainer/blob/master/Documentation.pdf